MIL-C-60051 (MU)
15 August 1964

# MILITARY SPECIFICATION

## COMPOSITION 86/14, RDX/WAX

### 1. SCOPE

1.1 Scope.-This specification covers one type of high explosive designated as Composition 86/14 RDX/Wax for use in ammunition.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, For Testing Purposes

MILITARY

MIL-R-398 - RDX
MIL-A-2550 - General Specification for Ammunition Except Small Arms Ammunition
MIL-W-20553 - Wax, Desensitizing (see 6.3)
MIL-I-45208 - Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1375

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DRAWINGS

ORDNANCE CORPS

F7548644 - Box, Packing for High Explosives. Assembly Details, Packing and Marking
F7548645 - Carton, Packing, Reusable-Collapsible for High Explosives, Assembly, Details, Packing and Marking

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publications

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revisions)".)

3. REQUIREMENTS

3.1 Material.-The composition shall consist of RDX crystals, complying with the requirements of specification MIL-R-398, completely coated with desensitizer, complying with the requirements of Specification MIL-W-20553 (see 6.3).

3.2 Insoluble particles.-Not more than five particles shall be retained on USS number 60 sieve, when tested as specified in 4.3.1.

3.3 Moisture.-The moisture content shall be 0.25 percent maximum (max.) when tested as specified in 4.3.2 or 4.3.3.



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3.4 Viscosity

3.4.1 The efflux viscosity of the composition shall not exceed 37 efflux seconds, max., when tested as specified in 4.3.4.

3.5 Composition.-The composition shall be as specified in Table I, when tested as in 4.3.5 or 4.3.6.

Table I

| | |
|---|---|
| RDX, percent | 86.0 plus or minus 2.3 |
| Desensitizer, percent | 14.0 plus or minus 2.3 |

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208. If due to the nature of the manufacturing process the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of Specification MIL-I-45208 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment,) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.



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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of one or more batches of the composition produced by one manufacturer, in accordance with the same specification, or same specification revision, under one



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continuous set of operating conditions. Each lot shall consist of that quantity of the composition that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. Desensitizer from one interfix number, from one manufacturer only.

b. RDX of one type only, from one interfix lot number, from one manufacturer only.

See 6.4 for further information.

The product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235 when applicable).

4.2.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans, in accordance with Standard MIL-STD-1235, may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.2.2.1 Wooden box or fiberboard carton, prior to closing (see Drawing (dwg.) F7548644, F7548645)

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Liner pierced or torn ..... | Visual | 01001 |
| 102. | Liner improperly closed ..... | Visual | 01002 |
| 103. | Foreign matter ..... | Visual | 01003 |
| Minor: None defined. | | | |

4.2.2.2 Sealed wooden box (see dwg. F7548644)

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Box damaged ..... | Visual | 02001 |
| 102. | Top improperly assembled ..... | Visual | 02002 |
| 103. | Strapping broken, or loose .... | Visual-Manual | 02003 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 1.50 percent | | |
| 201. | Nail protruding ..... | Visual | 02004 |
| 202. | Marking misleading or unidentifiable ..... | Visual | 02005 |
| 203. | Strapping improperly assembled. | Visual-Manual | 02006 |

4.2.2.3 Sealed fiberboard carton (see dwg. F7548645)

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 percent | | |
| 101. | Assembly torn or pierced ..... | Visual | 03001 |
| 102. | DOD symbol misleading or unidentifiable ..... | Visual | 03002 |
| 103. | Strapping broken or loose ..... | Visual-Manual | 03003 |
| Minor: | AQL 0.40 percent | | |
| 201. | Marking misleading or unidentifiable ..... | Visual | 03004 |

4.2.3 Determination of insoluble particles, moisture, viscosity and RDX and wax content. (see 3.2 to 3.4) Major defects

4.2.3.1 Testing.-Samples shall be selected from each batch of the composition in such number and amount as to insure that the samples shall be representative of the lot. These samples shall be mixed to form a composite sample before performing the tests specified in 4.3. If the composite sample fails to comply with any of the requirements specified herein (see 3.2 to 3.5), the lot shall be rejected. Failure to meet any of these requirements shall be considered a major defect. The tests shall be performed as specified in 4.3.

4.3 Test methods

4.3.1 Insoluble particles - Code No. 04001.-Determine the insoluble particles in accordance with Method 106.1 of Standard MIL-STD-650.

4.3.2 Determination of moisture

4.3.2.1 Conductimetric method - Code No. 05001.-Standard Method



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4.3.2.2 Apparatus.-The following apparatus is required:

a. A conductivity cell having platinum electrodes coated with platinum black, and mounted in a glass case. The cell constant of the conductivity cell shall be approximately 0.1 reciprocal centimeter.

b. A conductivity bridge, of the Wheatstone type, having a range from 10,000 to 1000 ohms, and a precision of 2 percent or better.

c. A glass stirrer of the propeller type.

d. A stirring motor.

e. A 200 ml. automatic pipette connected by means of a ground glass connection with a 5-gallon pyrex reservoir.

f. A 50 ml. automatic burette connected by means of a ground glass connection with a 5-gallon pyrex reservoir.

g. A wide mouth glass titration bottle of approximately 400 ml. capacity equipped with a tight fitting ground glass stopper.

h. A rubber stopper which fits the titration bottle, and which is provided with 3 holes, one for the conductivity cell, one for the stirrer, and one for the tip of the burette.

4.3.2.3 Assembly.-The apparatus shall be assembled as follows: The conductivity cell shall be inserted into the appropriate hole of the stopper. A short piece of glass tubing shall be placed into another hole of the rubber stopper, to act as a sleeve for the shaft of the stirrer. The internal diameter of the glass tubing used shall be only slightly greater than the diameter of the shaft of the stirrer. The rubber stopper assembly shall be clamped to a suitable support. The shaft of the stirrer shall then be passed through the sleeve in the rubber stopper and attached to the stirring motor. The stirring motor shall be provided with a suitable support. The burette shall be set up so that the tip of the burette passes completely through the third hole in the rubber stopper. The electric leads from the conductivity cell shall be attached to the proper binding posts of the conductivity bridge.

4.3.2.4 Solutions

4.3.2.4.1 Solution "A" (acetic acid-sulfuric acid).-A portion of 18.0 liters of approximately 99.9 percent glacial acetic acid shall be mixed with 85 ml. of approximately 96 percent sulfuric acid and 30 ml. of approximately 97.5 percent acetic anhydride in the 5-gallon pyrex reservoir, which is to be used with the 200 ml. automatic pipette. This solution shall have a blank



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titration value of not more than 3 ml. of solution B as determined in 4.3.2.5. If one blank titration of the solution exceeds 3 ml. of solution B, more acetic anhydride shall be added.

4.3.2.4.2 Solution "B" (acetic anhydride-acetic acid).-A portion of 2.0 liters of approximately 97.4 percent acetic anhydride shall be mixed with 15.0 liters of approximately 99.9 percent glacial acetic acid in the 5 gallon pyrex reservoir which is fitted with the 50 ml. automatic burette. This solution shall have a water equivalent value of approximately 0.02 gm. of water per ml., when standardized as specified in 4.3.2.5. The water equivalent value of the solution shall be adjusted if necessary, by adding more acetic anhydride or glacial acetic acid, the former increasing the value and the latter decreasing it.

4.3.2.5 Standardization of the solutions

4.3.2.5.1 Preferred procedures.-Two hundred ml. of solution "A" shall be pipetted into the titration bottle. All necessary precautions shall be taken to minimize absorption of moisture from the atmosphere during this, and the subsequent operations. A weighed portion of approximately 0.5 gm. of water shall be added to the bottle, and the bottle attached to the rubber stopper assembly. The stirring motor shall be started, and operated at such speed that the solid material (if any) remains in suspension, and the resistance of the solution shall be practically constant as measured with the conductivity bridge. The agitated solution shall be titrated with solution B, 0.5 ml. portions being added at a time, and the resistance determined after each addition. As solution B is added, the resistance of the solution being titrated increases to a max. at the endpoint and then decreases. The resistance values obtained shall be plotted on rectangular coordinates against the corresponding number of ml. of solution "B" added. A straight line shall be drawn through the two points on the plot just preceding the point of max. resistance value, and the line extended to intersect a similarly extended line drawn through the two pointson the plot just following the max. resistance value. The point of intersection of the two lines shall be considered as the end-point of the titration. The number of ml. of solution "B" corresponding to this end-point shall be recorded. A blank determination shall be made on a 200 ml. portion of solution A. The number of gm. of water equivalent to 1 ml. of solution "B" shall be calculated as follows:

E = W divided by V-A



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where:
E = gm. of water equivalent to a ml. of solution "B".
W = gm. of water added in standardization.
V = ml. of solution "B" required to titrate the specimen.
A = ml. of solution "B" required to titrate the blank.

4.3.2.5.2 Alternate procedure.-The following procedure for determination of the titration end-point may be used as an alternate to the procedure specified in 4.3.2.5.1. The progress of the titration shall be followed by adjusting the conductivity bridge reading, sufficiently above or below the actual resistance of the solution so that the shadow in the "eye" appears as a hairline. The reading of the bridge shall be increased uniformly so that the appearance of the shadow remains constant through the titration. The end-point is shown in the "eye" by a slight opening of the shadow if the reading of the bridge is just below the resistance of the solution, and by the fading or disappearance of the shadow if the reading of the bridge is just above the resistance of the solution.

4.3.2.6 Procedure.-A specimen of the sample shall be crushed to a particle size of approximately 3/4 inch or smaller. A weighed portion of approximately 50 gm. shall be transferred to the titration bottle. A 200 ml. portion of solution "A" shall be added, necessary precaution being taken to minimize absorption of moisture from the atmosphere by the solution during this operation. It shall be ascertained that none of the specimen remains on the ground surfaces, then the bottle shall be stoppered with the glass stopper, and the contents of the bottle agitated until all of the RDX is dissolved. The glass stopper shall be removed and the bottle attached immediately to the rubber stopper assembly. The solution shall be titrated and the number of ml. of solution "B" equivalent to the end-point of the titration determined in a manner similar to that used in the standardization of solution "B", as specified in 4.3.2.5. The percentage of moisture in the specimen shall be calculated as follows:

Percent moisture = 100 E (V-v) divided by W

where:
V = ml. of solution "B" required to titrate the specimen
v = ml. of solution "B" required to titrate the blank
E = gm. of water equivalent to 1 ml. of solution "B"
W = gm. of specimen



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4.3.3 Alternate Moisture Method (Karl Fischer Method) - Code No. 06002.-Determine the moisture content in accordance with Method 212.1 of Standard MIL-STD-650.

4.3.4 Determination of Viscosity (Efflux Method) - Code No. 07001.-The viscosity shall be determined in accordance with Method 212.1 of Standard MIL-STD-650 except that the determination shall be made at a temperature of 95° Centigrade.

4.3.5 Composition Determination (Dimethylformamide Method)

Wax Content - Code No. 08001
RDX Content - Code No. 09001

Place approximately 75 grams of the sample on a crushing board and crush. Transfer the crushed sample to U.S.S. sieves (#14 and #35) and shake. Use the material passing through the #14 sieve and retained on the #35 sieve for the determination which is depicted below:

Place a weighed portion of the crushed sample, 5.00 gms. to the nearest 0.1 milligram, into a 250 ml. beaker and add 175 ml. of dimethylformamide (DMF). Stir the solution for 30 minutes using a teflon coated magnetic stirring bar apparatus. Filter the solution through a tared medium porosity gooch crucible fitted with a 2.8 inch fiber glass filter pad. Place the crucible in a vacuum flask connected to a source of vacuum as an aid for the filtration. Transfer the insoluble residue from the beaker to the gooch crucible using three 10 ml. portions of DMF. Draw air through the gooch crucible until the crucible is dry of DMF. Wash the gooch crucible and contents with a 20 ml. portion of acetone. Place the gooch crucible in a vacuum oven and dry at 70° C. for one hour or until constant weight is obtained. Calculate the percentage of Wax and RDX as follows:

$$\text{Percent Wax} = \frac{100\ A}{W-(MW)}$$

$$\text{Percent RDX} = 100 - \text{Percent Wax}$$

where:
A = weight of insoluble matter, gms.
W = weight of sample, gms.
M = percent moisture in material expressed as a decimal (see 4.3.2).



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(NOTE: Run this determination in duplicate. Use the average of the two results if their values differ by less than 0.2 percent. If their values differ by greater than 0.2 percent, rerun the determination and use the average of the three values which are in closest agreement.)

4.3.6 Alternate Composition Determination (Benzene Extraction Method)

Wax Content - Code No. 10001
RDX Content - Code No. 11001

Place approximately 75 grams of the sample on a crushing board and crush. Transfer the crushed sample to U.S.S. sieves (#14 and #35) and shake. Use the material passing through the #14 sieve and retained on the #35 sieve for the determination which is depicted below:

Place a weighed portion of the crushed sample, 5.00 gms. to the nearest 0.1 milligram, into a 100 ml. beaker and add 40 ml. of benzene. Cover beaker with cover glass and place beaker with sample on a steam hot plate for exactly 30 minutes. Stir the solution in the beaker, during this 30 minutes, with a glass rod to assure that all the wax dissolves. Filter the solution through a preheated, tared medium porosity gooch crucible. Place the crucible in a vacuum flask connected to a source of vacuum as an aid for filtration. Transfer the insoluble residue from the beaker to the gooch crucible, while hot, with 3 to 4-10 ml. portions of benzene. Draw air through the gooch crucible and contents until the crucible is dry of benzene odors. Concurrently with the above operations run a blank using an accurately weighed sample (within 0.1 milligram) of approximately 4.3 grams of RDX to compensate for the RDX solubility in benzene. Determine the blank by performing the operations in exactly the same manner as the sample under test. Dry the crucible and contents for one hour at 100 degrees C., plus or minus 5 degrees C., cool in a desiccator, and weigh. Calculate the percentage of RDX and Wax as follows:

$$\text{Percent RDX} = \frac{100\ (A \text{ plus } B)}{W-(MW)}$$

Percent Wax = 100 - Percent RDX



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Where:
A = weight of insoluble matter, gms.
B = weight of RDX (gms.) lost in Blank determination
W = weight of sample, gms.
M = percent moisture in material expressed as a decimal (see 4.3.2)

(NOTE: Run this determination including blank in triplicate. Use the average of the three results if their values differ by less than 0.2 percent. If their values differ by greater than 0.2 percent, rerun the determination and use the average of the four values which are in closest agreement.)

## 5. PREPARATION FOR DELIVERY

### 5.1 Packing (see 6.1)

5.1.1 Level A.-The composition shall be packed and marked in accordance with dwg. F7548644.

5.1.2 Level B.-The composition shall be packed and marked in accordance with dwg. F7548645.

## 6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:
a. Title, number and date of this specification.
b. Level of packing protection required.

6.2 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 The type of wax used in the 86/14 composition should be restricted to Sunoco 8810, or some other wax determined to be suitable for use in this composition.

6.4 A lot (or batch), once prepared cannot be reworked without a request for waiver.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A775



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